

Hemant Kumar

Portfolio: hemantk.me
Github: github.com/HemantKumar01
Linkedin: linkedin.com/in/hemant-kumar-a6ba8815a/

Email: ytbhemant@gmail.com
Mobile: +91-8756839271

EDUCATION		
• Indian Institute of Technology (IIT), Patna	Patna, India	
• Bachelor of Technology - Computer Science and Engineering; CGPA: 9.09	Aug. 2023 - May 2027 (Expected)	
Courses: Algorithms, Artificial Intelligence, Discrete Mathematics, Computer Architecture, Entrepreneurship, Database		
• LBS Sr. Sec School, Kota, India	Kota, India	
• Senior School Certificate Examination (CBSE), Percentage: 96%	2023	
• Kendriya Vidyalaya, Nasirabad, India	Ajmer, India	
• Secondary School Examination (CBSE), Percentage: 98.4%	2021	

ACHIEVEMENTS	
• Rank 24 in NK Securities Research Hackathon for implied volatility prediction among 2095 entrants.	2025
• Selected among Top 100 projects for Google Solutions Challenge 2025, out of 64,000 participants.	2025
• Secured All-India Rank 26 at the ICPC Regionals 2024.	2024
• Rated Expert (max. rating 1621) on Codeforces (CF Handle: XGB)	2024
• Secured 2nd Position in Codepeak 2023 a prestigious one-month-long open-source event.	2023
• Secured 2552 rank (top 1%) in JEE Advanced exam among 250,000+ students	2023
• Secured 3216 rank (99.72%ile) in JEE Mains exam among 1.1 Million students.	2023
• INMO: Got selected for Indian National Math Olympiad by being top 135 students in India.	2023

EXPERIENCE	
• Droidrun GmbH	Remote
• Developer Relations Engineer	Oct. 2025 - Jan. 2026
◦ Designed and implemented a stealth ADB abstraction layer to bypass emulator bot-detection mechanisms, improving reliability of automated mobile interactions (merged PR #231)	
◦ Built and executed a scalable benchmarking framework to stress-test mobile-use agents using parallel asynchronous LLM-driven workflows, enabling performance and latency analysis under high concurrency. (Repository)	
◦ Integrated LLM token accounting and cost analysis into agent execution pipelines, providing step-level visibility into usage and optimizing inference costs (merged PR #142)	
◦ Developed end-to-end mobile automation pipelines using Droidrun, including WhatsApp, LinkedIn, and Twitter bots, demonstrating real-world agent orchestration across consumer applications.	
• Revspot.ai	Remote
• Software Developer Intern	Aug. 2024 - Oct. 2024
◦ Developed a profile fit model to predict lead conversion from a raw, highly skewed dataset by implementing oversampling techniques; achieved 97% accuracy and 80% ROC AUC .	
◦ Built a responsive, user-centric website , significantly enhancing user engagement and improving lead conversion rates.	
◦ Implemented JWT-based authentication using Auth0 , enabling secure, multi-organization access across 4 separate client environments .	

PROJECTS

- **TweetFire - Twitter Automation System** Source Code
Autonomous AI agents generating and posting viral Twitter content without APIs Nov. 2025 - Nov. 2025
 - **Tools & technologies:** Gemini, Google Trends, Android Automation, Python
 - Built an end-to-end **AI agent pipeline** to identify real-time trending topics using Google Trends and autonomously generate Twitter-ready content.
 - Integrated **Gemini-powered image and text generation** to create high-engagement, clickbait-style visuals and captions optimized for virality.
 - Implemented **Android device automation** to publish posts directly on Twitter, eliminating dependency on Twitter or Trends APIs.
- **SoulScript - AI Mental Health Therapist** Deployed Link — Source Code
AI mental health therapist that talks to you in realtime and helps you know about yourself. Apr. 2025 - Jun. 2025
 - **Tools & technologies:** Gemini, Google Agent Development Kit, Typescript, Flask
 - Developed a real-time conversational **AI avatar** with **<1s audio latency**, enabling seamless voice interactions using Gemini's **native audio APIs**.
 - Engineered a multi-agent system using **Google ADK** and Python Flask backend to dynamically generate therapeutic responses and track session progress across 1000+ concurrent users.
 - Implemented automated persona report generation via **Retrieval-Augmented Generation (RAG)** on psychological literature, enhancing therapy personalization for over 90% of users.
- **Two-Pass RISC Assembler and Emulator (C++)** Source Code
Under the guidance of Prof. Jimson Mathew Nov. 2024
 - Built a high-performance two-pass RISC assembler in C++ to convert assembly code into machine code.
 - Implemented symbol table generation, syntax/semantic validation, and label resolution in the first pass.
 - Developed an emulator to execute machine code with step-by-step register and memory state visualization.
- **OneIITP - Official Cross Platform App of IIT Patna** Play Store — App Store
Gained more than 2000 active users within a week Aug. 2024 - Jan., 2025
 - **Tools & technologies:** React Native, Firebase
 - Built a unified mobile app for IIT Patna students, addressing a wide range of daily campus needs through a clean, intuitive interface, supported on **both Android and IOS**.
 - Reduced attendance proxy by 96% through a custom-built, **encrypted QR-based anti-proxy system** that **changes QR every second** to prevent sharing. ([Attendance System Link](#))
 - **Key Features:** Live bus schedules, blogs, class timetable, attendance tracker, institute marketplace, events, quick links, emergency contacts, food outlets, and an admin mode.
- **Ark Wallet (Bitcoin)** Source Code
A Bitcoin wallet supporting off-chain and on-chain payments via the Ark network Apr 2025
 - **Tools & technologies:** Rust, Ark Protocol, Nigiri (Bitcoin regtest environment)
 - Built a feature-rich wallet that enables users to **create/manage Ark wallets**, generate **on-chain/off-chain addresses**, and **send/receive funds** seamlessly across the Ark network.
 - Implemented **faucet** functionality to request Bitcoin on regtest from **Nigiri**, streamlining the local development and testing process.
 - Added settlement capability to move funds between **off-chain and on-chain wallets**, simulating real-world Bitcoin transaction flow and ensuring protocol compliance.
 - Demonstrated secure handling of digital assets within a **regtest environment**, supporting robust integration testing and iteration.

SKILLS SUMMARY

- **Languages:** Python, C, C++, Rust, JavaScript, Typescript, SQL
- **Frameworks:** Scikit-learn, TensorFlow, PyTorch, Flask, NodeJS
- **Tools:** Docker, Git, MongoDB, MySQL
- **Platforms:** Linux, Web, Android, Google Cloud Platform
- **Soft Skills:** Teamwork, Leadership, Communication, Analytical Solving

POSITION OF RESPONSIBILITY

- **Overall Lead,** Google Developer Groups on Campus, IIT Patna Aug. 2024 – Present
- **Sub-Coordinator,** NJACK – Programming Club, IIT Patna Jun. 2024 – Dec. 2024